

Byoungyun Gong

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SUMMARY

Biomedical Engineering M.S. student with a biotechnology background and hands-on experience in molecular biology, including experimental troubleshooting, physiological testing, and fluorescence imaging. Developing computational and quantitative skills in Python-based data analysis and machine learning for applications in biomedical research and healthcare technologies.

EDUCATION

Arizona State University

M.S. Biomedical Engineering

Tempe, AZ

Aug 2026 – Present

Yonsei University

B.S. Bioengineering & Biotechnology | GPA: 3.56/4.30

Seoul, South Korea

Mar 2021 – Aug 2025

Relevant Coursework: Molecular Biology, Regenerative Biotechnology, Biodata and Information Processing, Mathematics for Biotechnology (ODEs, Fourier Series, Probability), Introduction to AI and Its Application, SW Programming

TECHNICAL SKILLS

Programming & Data: Python, NumPy, pandas, Matplotlib, scikit-learn, Keras, Jupyter Notebook, Google Colab

Molecular & Cellular: DNA Extraction, PCR, Genotyping, Gel Electrophoresis, Mammalian Cell Culture, Whole-Mount Immunofluorescence

Biomedical & Imaging: Auditory Brainstem Response (ABR), Cochlear Microdissection, Confocal Microscopy

Computational Biology: DNA/RNA Sequence Analysis

RESEARCH EXPERIENCE

Undergraduate Research Intern

Yonsei University College of Medicine

Mar 2025 – Jun 2025

Seoul, South Korea

- Investigated molecular mechanisms of *SLC26A4*-associated hearing loss using *IVS7-2A>G* and *hH723R* transgenic mouse models
- Conducted Auditory Brainstem Response (ABR) testing and interpreted waveforms across 6–30 kHz to determine hearing thresholds and characterize auditory phenotypes in *hH723R* mutant mice
- Troubleshoot faint and non-specific PCR amplification by optimizing a touchdown PCR genotyping protocol
- Performed cochlear microdissection and whole-mount immunofluorescence to evaluate hair-cell morphology and protein localization
- Genotyped 150 transgenic mouse samples via DNA extraction and PCR for mutant identification

Undergraduate Research Intern

Yonsei University

Jul 2023 – Aug 2023

Seoul, South Korea

- Extracted genomic DNA from five cancer cell lines and performed PCR amplification targeting KRAS mutation hotspots for Sanger sequencing
- Identified KRAS point mutations and distinguished heterozygous and homozygous variants from Sanger sequencing results

PROJECTS

Regression & Image Classification, Class Project

Fall 2024

- Trained and evaluated a multiple linear regression model on the scikit-learn diabetes dataset using RMSE and R^2
- Tuned hyperparameters of a provided MLP model for Fashion-MNIST image classification and evaluated its performance using Keras

Biotechnology Laboratory, Class Projects

Spring 2023

- Fabricated hyaluronic acid-based dissolving microneedles using controlled dispensing and centrifugation for transdermal drug delivery
- Synthesized calcium-crosslinked alginate hydrogels for 3D cell encapsulation and assessed cell viability from Live/Dead fluorescence images

ACTIVITIES

Mentor & Presenter

Severance Hospital Pediatric Cochlear Implant Information Session

May 2026

Seoul, South Korea

- Developed and delivered a 10-minute invited presentation to 30+ attendees, primarily parents of children with hearing loss, on accessibility and my patient-to-researcher perspective on hearing science